TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	79.3 ± 3.47
Correlation Stability	87.7 ± 2.15
Distribution Consistency	86.0 ± 2.49
Entropy Preservation	88.9 ± 1.70
Mean Z Score Drift	0.2 ± 0.04
Categorical Fidelity	82.6 ± 0.66
Privacy Risk	49.1 ± 1.37
Data Density	100.0 ± 0.01
Coverage Score	21.0 ± 5.20
Feature Accuracy	77.5 ± 2.06
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	41.687	-3.059	6.8%	OK
TB	3.299	1.577	-1.722	52.2%	Drift
DB	1.486	0.626	-0.860	57.8%	Drift
Alkphos	290.576	208.004	-82.572	28.4%	Drift
Sgpt	80.714	38.597	-42.117	52.2%	Drift
Sgot	109.911	39.401	-70.510	64.2%	Drift
TP	6.483	6.818	+0.335	5.2%	OK
ALB	3.142	3.560	+0.418	13.3%	Warn
A/G Ratio	0.947	1.078	+0.131	13.9%	Warn
Selector	1.286	1.442	+0.156	12.1%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	79.3 ± 3.47
Correlation Stability (*2)	87.7 ± 2.15
Distribution Consistency (*3)	86.0 ± 2.49
Entropy Preservation (*4)	88.9 ± 1.70
Mean Z Score Drift (*5)	0.2 ± 0.04
Categorical Fidelity (*6)	82.6 ± 0.66
Privacy Risk (*7)	49.1 ± 1.37
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	21.0 ± 5.20

Feature Accuracy (*10)	77.5 ± 2.06
Novelty Score (*11)	0.4 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6572649572649574	0.5247863247863248	0.6384615384615384	-0.01880341880341878	
F1	0.7600051231540046	0.6300415013976866	0.7529269425087068	-0.00707818064529772 7	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.5957264957264957	0.6709401709401711	-0.01538461538461538 5	
F1	0.7971299319768493	0.7063439997721824	0.7875543926384074	-0.00957553933844196 3	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7102564102564103	0.5717948717948719	0.7136752136752136	0.00341880341880340 67	
F1	0.8123278049093333	0.6833590377633011	0.8179005805497883	0.00557277564045515 2	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.5683760683760685	0.6923076923076923	-0.01367521367521368 2	
F1	0.8128002697439015	0.6726344470836908	0.8041104456044448	-0.00868982413945676	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6794871794871795	0.5435897435897437	0.6982905982905983	0.01880341880341881 3	
F1	0.7871273446765535	0.6459850558400405	0.7964111235272145	0.00928377885066055 2	

Model: XGB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5700854700854701	0.6794871794871795	-0.023076923076923096	
F1	0.7997903571549948	0.6722160078162174	0.7810542806378227	-0.018736076517172028	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6803418803418804	0.7068376068376068	0	
F1	0.8279091624529537	0.8073360251006456	0.8279091624529537	0	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7153846153846154	0.5367521367521368	0.6829059829059829	-0.03247863247863249	
F1	0.8154758491723733	0.6499471627168695	0.7969767056029636	-0.018499143569409682	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5427350427350428	0.5623931623931624	0.00341880341880344	
F1	0.5587432119380917	0.6398888395425002	0.5676457472805198	0.008902535342428141	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5572649572649573	0.7008547008547008	0.026495726495726502	
F1	0.7821343746683586	0.6644252451139361	0.8092584275674353	0.027124052899076624	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	64	Female	0.8	0.2	178	17	18	6.3	3.1	0.9
2	38	Female	0.7	0.1	152	90	21	7.1	4.2	1.4
3	46	Male	9.4	5.2	268	21	63	6.4	2.8	0.8
4	58	Male	0.8	0.2	123	56	48	6.0	3.0	1.0
5	42	Male	0.8	0.2	198	29	19	6.6	3.0	0.8
6	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
7	29	Female	0.6	0.1	150	25	20	6.5	3.4	1.1
8	54	Male	2.2	1.0	200	40	35	7.0	4.0	1.2
9	41	Female	1.5	0.7	180	30	28	6.8	3.5	1.0
10	37	Male	0.9	0.3	190	22	25	7.2	3.6	1.3
11	62	Female	0.8	0.2	210	50	30	6.4	3.2	0.9
12	48	Male	3.0	1.5	250	60	40	7.5	4.1	1.5
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	35	Male	0.8	0.2	279	20	25	7.2	3.2	0.8
2	30	Male	1.6	0.4	332	84	139	5.6	2.7	0.9
3	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
4	60	Male	0.8	0.2	286	21	27	7.1	4.0	1.2
5	46	Male	3.3	1.5	172	25	41	5.6	2.4	0.7
6	32	Female	0.6	0.1	176	39	28	6.0	3.0	1.0
7	29	Female	0.9	0.3	210	45	50	6.5	3.0	0.8
8	41	Male	1.2	0.5	300	60	70	7.0	3.5	1.0
9	36	Female	0.8	0.2	180	50	55	6.2	2.8	0.7
10	50	Male	2.5	1.1	400	80	90	7.5	3.8	1.2
11	62	Female	1.1	0.4	250	70	75	6.8	3.2	0.9
12	28	Male	0.7	0.1	190	30	35	5.5	2.5	0.6
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	51	Male	0.8	0.2	230	24	46	6.5	3.1	nan
2	54	Male	2.2	1.2	195	55	95	6.0	3.7	1.6
3	66	Male	0.8	0.2	165	22	32	4.4	2.0	0.8
4	52	Male	0.9	0.2	156	35	44	4.9	2.9	1.4
5	30	Female	0.7	0.2	63	31	27	5.8	3.4	1.4
6	53	Female	0.7	0.1	182	20	33	4.8	1.9	0.6
7	29	Female	0.9	0.3	150	25	30	6.0	3.2	0.8
8	37	Male	1.2	0.5	180	45	50	7.0	3.5	0.9
9	48	Female	0.8	0.2	200	35	40	6.5	3.1	0.7
10	54	Male	1.5	0.6	220	50	60	7.2	3.8	1.0
11	62	Female	0.7	0.1	170	30	35	5.8	2.9	0.8
12	45	Male	2.1	1.0	250	60	70	7.5	3.6	1.1
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	70	Male	1.4	0.6	146	12	24	6.2	3.8	1.58
2	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
3	45	Male	1.7	0.8	315	12	38	6.3	2.1	0.5
4	42	Female	2.3	1.1	292	29	39	4.1	1.8	0.7
5	65	Male	4.9	2.7	190	33	71	7.1	2.9	0.7
6	42	Female	0.8	0.2	195	18	15	6.7	3.0	0.8
7	34	Female	1.2	0.4	210	25	30	6.5	3.5	1.0
8	58	Male	0.9	0.3	175	40	50	7.2	3.8	1.2
9	45	Female	2.5	1.0	220	30	35	6.8	3.2	0.9
10	67	Male	3.1	1.5	300	50	70	7.5	4.0	1.1
11	52	Female	0.7	0.2	190	20	25	5.9	2.5	0.8
12	39	Male	1.8	0.9	250	35	45	6.3	3.0	1.0
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	0.7	0.2	192	18	15	7.4	4.2	1.3
2	42	Male	6.8	3.2	630	25	47	6.1	2.3	0.6
3	48	Male	1.4	0.6	263	38	66	5.8	2.2	0.61
4	58	Male	1.7	0.8	188	60	84	5.9	3.5	1.4
5	35	Male	0.9	0.2	190	25	20	6.4	3.6	1.2
6	18	Male	0.6	0.2	538	33	34	7.5	3.2	0.7
7	34	Female	1.2	0.5	210	30	40	6.5	3.1	1.05
8	29	Male	0.9	0.3	180	22	25	7.0	3.8	1.2
9	45	Female	2.5	1.0	300	40	50	7.2	4.0	1.15
10	52	Male	0.7	0.2	250	35	45	6.8	3.5	1.1
11	39	Female	1.8	0.9	220	28	32	6.0	2.9	1.03
12	61	Male	3.0	1.5	400	50	60	7.5	4.2	1.18
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3
2	26	Male	2.0	0.9	157	54	68	6.1	2.7	0.8
3	24	Male	3.3	1.6	174	11	33	7.6	3.9	1.0
4	22	Male	0.6	0.2	202	78	41	8.0	3.9	0.9
5	37	Male	0.8	0.2	147	27	46	5.0	2.5	1.0
6	65	Female	0.7	0.2	182	23	28	6.8	2.9	0.7
7	29	Female	1.2	0.5	150	35	40	6.5	3.0	1.2
8	34	Male	0.8	0.3	200	45	55	7.0	3.5	1.0
9	42	Female	2.1	0.9	220	50	60	6.8	3.2	1.1
10	55	Male	3.5	1.5	300	60	70	7.5	4.0	1.3
11	38	Female	0.6	0.2	180	30	35	5.9	2.8	1.0
12	47	Male	1.9	0.7	250	55	65	6.2	3.1	1.1
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	47	Male	0.9	0.2	265	40	28	8.0	4.0	1.0
2	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
3	66	Male	0.8	0.2	165	22	32	4.4	2.0	0.8
4	38	Male	1.7	1.0	180	18	34	7.2	3.6	1.0
5	48	Female	0.8	0.2	218	32	28	5.2	2.5	0.9
6	60	Male	2.3	0.6	272	79	51	6.6	3.5	1.1
7	34	Female	1.2	0.4	210	25	30	7.5	3.5	1.1
8	55	Male	0.9	0.2	180	22	20	6.0	3.0	0.9
9	42	Female	1.5	0.5	250	30	35	8.0	4.0	1.2
10	29	Male	0.7	0.1	160	18	15	5.5	2.5	1.0
11	38	Female	2.0	0.8	220	28	25	7.0	3.8	1.1
12	61	Male	3.5	1.5	300	40	50	9.0	4.5	1.3
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	62	Male	0.7	0.2	162	12	17	8.2	3.2	0.6
2	51	Female	0.9	0.2	280	21	30	6.7	3.2	0.8
3	45	Female	0.9	0.3	189	23	33	6.6	3.9	nan
4	62	Male	5.0	2.1	103	18	40	5.0	2.1	1.72
5	31	Female	0.8	0.2	158	21	16	6.0	3.0	1.0
6	42	Female	0.5	0.1	162	155	108	8.1	4.0	0.9
7	29	Female	1.2	0.4	210	30	25	6.5	3.2	0.9
8	54	Male	0.8	0.3	180	40	35	7.0	3.5	1.0
9	39	Female	2.5	1.1	250	50	45	6.8	3.4	0.8
10	47	Male	1.0	0.5	220	36	29	7.2	3.1	1.1
11	34	Female	0.9	0.2	190	25	20	6.4	3.0	0.7
12	61	Male	3.0	1.5	300	70	55	7.5	3.6	1.2
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
2	34	Male	6.2	3.0	240	1680	850	7.2	4.0	1.2
3	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
4	37	Male	0.8	0.2	195	60	40	8.2	5.0	1.5
5	32	Male	12.7	6.2	194	2000	2946	5.7	3.3	1.3
6	37	Male	0.7	0.2	235	96	54	9.5	4.9	1.0
7	29	Female	1.2	0.5	150	30	35	6.5	3.0	0.8
8	45	Male	3.5	1.2	200	50	60	7.0	4.0	1.2
9	54	Female	0.9	0.3	180	25	30	6.0	2.8	0.7
10	38	Male	2.1	0.9	220	40	45	7.5	3.5	0.9
11	62	Female	1.8	0.6	160	35	50	6.8	3.2	1.0
12	27	Male	0.5	0.2	140	20	25	5.5	2.5	0.6
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	40	Male	3.5	1.6	298	68	200	7.1	3.4	0.9
2	66	Female	2.9	1.3	168	21	38	5.5	1.8	0.4
3	75	Male	6.7	3.6	458	198	143	6.2	3.2	1.0
4	15	Male	0.8	0.2	380	25	66	6.1	3.7	1.5
5	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1
6	60	Male	0.5	0.1	500	20	34	5.9	1.6	0.37
7	34	Female	1.2	0.5	210	45	50	6.5	3.2	0.8
8	29	Male	0.9	0.3	150	30	40	7.0	3.5	1.0
9	55	Female	2.5	1.0	300	50	60	7.2	3.8	0.9
10	48	Male	3.0	1.5	400	70	80	6.8	3.0	0.7
11	62	Female	1.5	0.6	220	40	55	6.0	2.9	0.9
12	37	Male	2.0	0.8	180	35	45	7.5	3.6	1.1
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3